Assessing Ranking Fidelity of Competition Structures via Weighted Mutual Information

Abstract

We evaluate multiple competitions structures to assess their effectiveness in accurately reproducing the true underlying ranking of participating teams. Using repeated simulations for each structure, we compare teams' true performance ranks to their simulated outcomes. To quantify the accuracy of each format, we calculate weighted mutual information between true and simulated rankings across a large number of replicates. "Salient" weights are applied to look at only the top i out of n total teams/competitors in a given competition structure. This information-theoretic approach allows us to objectively compare formats and identify which structures most effectively preserve rank order. Results are visualized through comparative plots, providing clear insights into the trade-offs and strengths of each competition design.

| Keywords: tournament structures, mutual information

1 Introduction

There are hundreds of different competitions structures that dictate how a set of teams compete against each other to determine an overall ranking. Common structures include bracket tournaments where if a team wins their competition they advance to the next round. Bracket tournaments can be run as single elimination where competitors advance or are eliminated based on a single match (e.g. NCAA College Basketball Tournament, National Football League (NFL) playoffs, major tennis tournaments) or where each round is contested as a best of k series (e.g. $k = 7$: National Basketball Association (NBA) playoffs, National Hockey League (NHL) playoffs, k varies by round: Major League Baseball (MLB), Women's National Basketball Association (WNBA)) and the series winner advance while the series loser is eliminated. Additional styles of the bracket tournaments include variations where competitors are not eliminated after a single loss such as double elimination tournaments where competitors continue until they have lost twice. Within double elimination tournament, there are several variations. For instance, the NCAA College Wrestling tournament is a double elimination tournament, however, if a competitor loses a match they fall into the consolation

bracket and the best they can finish after a loss before the finals is 3rd place making this not a "true" double elimination format. Olympic judo uses a variant of this system with single elimination until the quarterfinals followed by a consolation bracket consisting of the four quarterfinal losers who still have a chance for a bronze medal[1].

"True" double elimination - a bracket tournament where a competitor can still finish ranked first as long as a competitor has not lost two games - is used in NCAA baseball and softball for certain rounds in their postseason. Additional variants of double elimination include double elimination brackets with a "true third" match which includes an additional match contested between the competitors ranked 2nd (loser of the finals) and 3rd (winner of the consolation bracket) at the end of the tournament *if they have not already competed against each other*. There is also a variation of double elimination where only those competitors who have lost to a competitor that makes the finals are placed into the consolation bracket and given the chance to compete for third place. This style of competition is used in Olympic wrestling (both freestyle and Greco-Roman)[2].

In bracket style competitions, only a handful of the match-ups that are possible actually occur. When every possible competitive pairing occurs, this is referred to as a round-robin structure where every competitor plays every other competitor the same number of times. Single round-robin, where was possible match-up occurs exactly once, is used in, for examples, the group stage of the FIFA World Cup to create a ranking to determine which teams advance to the next round. Double round-robin, where each possible pairing occurs twice, is used in association football in English Premier League (EPL) and in chess during the FIDE candidates tournament (a tournament that decides who will challenge the world champion). For the sake of brevity, we review only a small number of possible competition structures, but the interested reader can find a more complete list of competition structures in Devriesere and Goossens (2025), Appleton (1995) and Csato (2021).

Daniels (1969)

A natural question of interest after reviewing different types of competition structures is: which competition structure is best? As with most complicated questions, the answer is "it depends". One commonsense conceptualization for the best competition structure is that the final rankings produced by a competition structure should closely match the true rankings of the strength of the competitors. That is, a good competition structure should rank the true best team first, the true second best team second, and so on, until finally ranking the true worst team last. We refer to this property of a competition structure as the efficacy of the structure (Lasek and Gagolewski (2018), Sziklai, Biró, and Csató (2022)). If one

[1] in Olympic judo they award two bronze medals to the two competitors who each win their side of the consolation bracket

[2] As in Olympic judo, Olympic wrestling also award two bronze medals to the winners of each side of the consolation bracket

were only interested in correctly ranking the true best competitor as first, with the ranks of the remaining competitors viewed as inconsequential, we refer to this here are effectivity (Glenn (1960)). While there are many competition structures where efficacy and effectivity are useful measures[3], there are many other competitions where the correct ranking of more than one competitor, but not all the competitors is of import. For instance, in Olympic competition, the goal of a competition structure is to correctly rank the top 3 competitors to award gold, silver, and bronze medals to[4]. In many American sports, for instance the NFL, the goal of the competition structure of the regular season is to rank the top 7 teams in each conference to choose the playoff teams with the ranking of the teams 8 through 16 in each conference inconsequential in determining a Super Bowl champion[5]. We coin the term "salient ranks" to refer to the ranks of a competition structure that matter. That is the ranks 1, 2, and 3 in Olympic competition are the salient ranks in that competition structure. In this notation, efficacy is a measure with with salient ranks 1 through n where n is the number of competitors, and effectivity is a measure where 1 is the only salient rank.

If the only goal of a competition structure was high efficacy of the salient ranks, the best way to achieve this would be to play each pair of possible match ups of competitors as many times as possible. Of course, this is impossible due to many factors including time constraints, cost constraints, and wear and tear on competitors in high impact sports (e.g. boxing, American football, etc.). In addition, competitions where the true best team is guaranteed or nearly guaranteed to win may be less appealing to fans and may drive down demand as discussed in Johnson and Fort (2022).

In light of this, many different competition structures have been proposed, several of which are mentioned previously. Given these many competition structures, ideally there would be a way comparing the performance of different competition structures to each other.

Here, we propose a novel metric for evaluating different competition structures using an information theoretic approach. As an analogy, we view the results of a tournament as a transmitted message, and we use information theory to assess how accurately the message (i.e. the tournament results) matches the true transmission (i.e. the true ranking of competitors).

Specifically, we use weighted mutual information with a weights based on squared error loss[6] between the ranks generated by a competition structure and the true ranks of the competitors. Additionally, we include an indicator vector to specify the salient ranks, allowing us to

[3]efficacy is a useful measure for the EPL, for example, as the ranking of the teams matters at both the high end (i.e. entry into special tournaments) and the low end (i.e. relegation status) and effectivity is a useful measure for the NCAA College Basketball tournament (i.e. main goal is to crown a champion)

[4]Some sports award two bronze medals (e.g. wrestling, judo, boxing, tae kwon do)

[5]The ranking of teams outside of the playoffs does matter for draft purposes though

[6]other loss functions can easily be incorporated into our method

compare different competition structures with different choices for salient ranks.

The remainder of this manuscript presents the mathematical foundation of the proposed measure followed by a simulation study estimating out proposed measure for different types of competition structures with varying choices of salient ranks. We close by offering a summary of the work and ideas for future directions.

2 Methods

Consider a set of n teams indexed from $i = 1, 2, \cdots, n$ each with an associated true strength parameter $\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_n)$ such that $(\theta_1 \geq \theta_2 \geq \ldots \geq \theta_n)$. Next, let $r(\boldsymbol{\theta})$ be the indexes of $\boldsymbol{\theta}$ when θ is sorted from largest to smallest so that $r(\boldsymbol{\theta}) = \{1, 2, \ldots, n-1, n\}$ (Note: While we allow for two values of θ to be equal to each other, we choose to assign unique ranks to each θ_i). Next define $T(\phi, s)$ as a tournament with schedule ϕ and seeding structure s. ϕ contains all the information about the scheduling of teams, which could be fully known prior to the tournament (*e.g.* round robin) or determined as the tournament progresses (*e.g.* single elimination tournament). We then let $\hat{r}(T(\phi, s), \boldsymbol{\theta})$ be an n-dimensional vector-valued random variable that gives the results of a tournament as a vector of the indexes of $\boldsymbol{\theta}$. The number in the first position of the vector $\hat{r}(T(\phi, s), \boldsymbol{\theta})$ indicates the index of the team that finished first, the number in the second position of the vector $\hat{r}(T(\phi, s), \boldsymbol{\theta})$ indicates the index of the team that finished second, and so on. (Note: We do allow for ties in $\hat{r}$.)

For example, in a $n = 4$ team tournament, $\hat{r}(T(\phi, s), \boldsymbol{\theta}) = (3,\ 2,\ 1,\ 4)$ indicates that the team with index 3 (i.e. the true third best team) won the tournament, the true second best team finished second, the true best team finished third and the true 4th team finished 4th in that particular tournament. If $\hat{r}(T(\phi, s), \boldsymbol{\theta}) = (1,\ 2,\ 3,\ 4)$, this means that the random outcome of the tournament was the same as the true ordering of the teams.

As noted above we allow for ties in $\hat{r}$. Continuing the above example, if $\hat{r}(T(\phi, s), \boldsymbol{\theta}) = (3,\ 2,\ 1,\ 3)$ this indicates that the teams that are truly ranked 1st and 4th tied for 3rd.

We need a transition: If one views the true ranking of the teams in a tournament as a message to be sent to a receiver and the outcome of the tournament as a message that is received, we can measure the "goodness" of a tournament in terms of its ability to accurately transmit the true ranking. We can then use concepts from information theory to assess the ability of a tournament to correctly rank teams. Specifically, we start with the concept of mutual information (Guiasu (1977)).

For two random variables X and Y mutual information is defined as:

$$I(X;Y) = \sum_{y \in Y} \sum_{x \in X} p(x, y) \log \frac{p(x, y)}{p(x)\, p(y)}$$

In our setting, we replace X and Y with $r(\boldsymbol{\theta})$ and $\hat{r}(T(\phi, s), \boldsymbol{\theta})$, and we could seek to estimate $I(r(\boldsymbol{\theta}), \hat{r}(T(\phi, s), \boldsymbol{\theta}))$. For simplicity, we drop the arguments from the functions r and $\hat{r}$ for ease of exposition.

In order to compute mutual information, marginal distributions of r and $\hat{r}$ are required. Without knowing r, all possible permutations of $\hat{r}$ are equally likely so we set $p(\hat{r}) = \frac{1}{n!}$. By a similar argument we set $p(r) = \frac{1}{n!}$.

we set . By a similar argument we set $p(\hat{r}) = \frac{1}{n!}$.

We define the mutual information of r and $\hat{r}$ to be:

$$I(r, \hat{r}) = \sum_{r} \sum_{\hat{r}} p(r, \hat{r}) \log \frac{p(r, \hat{r})}{p(r)p(\hat{r})} = n! \sum_{\hat{r}} p(r, \hat{r}) \log \frac{p(r, \hat{r})}{(\frac{1}{n!})^2}$$

While there are $n!$ different permutations for the result of r, we don't need to sum across these as any specific choice of r is just an arbitrary labeling of the true strength parameters vector $\boldsymbol{\theta}$. So given r, the distribution of $\hat{r}$ is the same up to the labeling. Therefore, we only need to consider a single permutation of r, we compute the the quantity inside the summation, and then multiple by $n!$ (effectively summing across r).

In order to compute this quantity, we need to compute $p(r, \hat{r})$. This probability is found by calculating the probability of a given permutation of $\hat{r}$ and r is assumed to be the permutation from $\{1, 2, \ldots, n\}$. We estimate $p(r, \hat{r})$ empirically through simulation.

However, using mutual information in this form does not suit our needs in this setting. The problem is that mutual information in this form will yield high values of mutual information when the output from the tournament $\hat{r}$ is highly consistent *even if the ranking from the tournament is incorrect* (*i.e.* the rankings produced by the tournament are very different from the true rankings). As an example, if $n = 4$ and the true order of $\boldsymbol{\theta}$ is $r = \{1, 2, 3, 4\}$ and $p(\hat{r} = \{4, 3, 2, 1\}) = 1$ will be the same mutual information as when $r = \{1, 2, 3, 4\}$ and $p(\hat{r} = \{1, 2, 3, 4\}) = 1$ and in both of these cases the mutual information will be maximized at:

$$I(r, \hat{r}) = n! \sum_{\hat{r}} p(r, \hat{r}) \log \frac{p(r, \hat{r})}{\left(\frac{1}{n!}\right)^2} = n! log((n!)^2)$$

and for the specific case when $n = 4$ would be:

$$4! log_2(4!^2) = 24 * log_2(24^2) = 220.0782$$

3 Weighted Mutual Information

In order to alleviate this problem, we instead consider *weighted* mutual information (Guiasu (1977)). We want to give more weight to permutations from $\hat{r}$ that are "closer" to r. Essentially, this problem comes down to choosing a metric that measures the closeness between the vectors $\hat{r}$ and r. One then gives more weight to combinations of $\hat{r}$ and r that are closer and less weight to combinations that are far apart. A natural choice for distance between vectors the sum of squared deviations between the two vectors. Specifically, we define the loss function between $\hat{r}$ and r to be $l(r, \hat{r}) = \sum_{i=1}^{n}(\hat{r}_i - r_i)^2$. Then, since small losses should be associated with large weights, we focus on the recipricol of this loss function.

In general, any loss function $l(r, \hat{r})$ can be used to define the weighting function as follows:

$$w(r, \hat{r}) = \begin{cases} \frac{1}{l(r,\hat{r})}, & r \neq \hat{r} \\ 1, & r = \hat{r} \end{cases}$$

Note that in order to avoid dividing by 0, we define the weight to be 1 when r and $\hat{r}$ are exactly equal.

Specifically, when squared error loss is chosen, one gets the following weight function:

$$w(r, \hat{r}) = \begin{cases} \frac{1}{\sum_{i=1}^{n}(\hat{r}_i - r_i)^2}, & r \neq \hat{r} \\ 1, & r = \hat{r} \end{cases}$$

An alternative to defining weight this way is to consider looking at the correlation between $\hat{r}$ and r. Since, $\hat{r}$ and r are already ranks, the Pearson and Spearman correlation coefficients will be computationally exactly the same in this setting if that was the measure that someone chose to use.

Spearman correlation can be written as follows:

$$\rho_s = 1 - \frac{6\sum_{i=1}^{n}(\hat{r}_i - r_i)^2}{n(n^2 - 1)}$$

,

which means that Spearman correlation is simply (one minus) a scaling of squared error loss. Since $\rho_s \in [-1, 1]$, but we desire weights $w(\hat{r}, r) \in [0, 1]$, we define $w(\hat{r}, r) = \frac{(\rho_s + 1)}{2}$ guaranteeing weights in the desired range.

While squared error loss and Spearman/Pearson correlation work well, in this setting we argue that Kendall's τ correlation is the most appropriate for weighting due to its interpretation and how this related to competitions. One way of interpreting Kendall's τ is that it's

a scaling of the number of adjacent swaps required to go from one permutation of ranks to another. Recall that Kendall's τ is defined as:

$$\rho_\tau = 1 - \frac{2 \times \text{\# discordant pairs}}{\binom{n}{2}}$$

. To convert ρ_τ back to the range between $[0, 1]$, which we want for weighting purposes, we subtract 1 and divide by 2 giving us $\frac{\rho_\tau + 1}{2}$. This then becomes:

$$w(\hat{r}, r) = \frac{\rho_\tau + 1}{2} = 1 - \frac{\text{\# discordant pairs}}{\binom{n}{2}}$$

and the number of discordant pairs is equivalent to the number of adjacent swaps needed to get the order of $\hat{r}$ correct.
As an example, consider $r = (1, 2, 3, 4)$ and $\hat{r} = (1, 3, 4, 2)$. This requires two adjacent swaps to get $\hat{r}$ to equal r (i.e swap 4 with 2 and then 3 with 2). The Kendall's τ value between these two vectors is $\frac{1}{3}$. We would then define the weight between the two vectors to be $(\rho_\tau + 1)/2 = \frac{2}{3}$. This is exactly equivalent to $1 - \frac{\text{\# adj swaps}}{\binom{n}{2}}$.

Once an appropriate weight function $w(r, \hat{r})$ is chosen, we get the following formula for weighted information:

$$I_w(r, \hat{r}) = \sum_r \sum_{\hat{r}} w(r, \hat{r}) p(r, \hat{r}) \log \frac{p(r, \hat{r})}{p(r)p(\hat{r})}$$

Tiered Ranking

This framework was developed to assess the efficacy of the full ranking of the top k teams generated by different competition formats. However, it may be of interest to have a competition structure that simply selects the top k teams in any order. Continuing with the Olympics example, if an organizer only cares about the top 3 teams earning medals *in any order* (i.e. $\hat{r} = \{1, 2, 3, 4\}$ and $\hat{r} = \{3, 2, 1, 4\}$ are equivalent), our framework can easily handle this modifying $\hat{r}$ (and r?) so that the top k elements are ordered correctly and tied. That is, for the Olympic example, if k was 3 and $\hat{r} = \{3, 2, 1, 4\}$, we convert all the elements of $\hat{r} \leq k$ to 1 (i.e. $\hat{r}$ becomes $\{1, 1, 1, 4\}$), and then we proceed with the exact same formula. We refer to this as tiered ranking.

This concept can be extended to multiple tiers, which is applicable to a competition structure such as the premier league where the top 4 teams qualify for the Champions league and the bottom 3 teams are relegated. While this straight foward to compute, we do not evaluate scenarios such as this in this article.

4 Salient Weights (and ranks)

In addition to the the weighting function w, we defined a second set of weights that we refer to as the "salient weights" and denote as the vector m. This set of weights is used to indicate which ranks have been determined to salient (as previously described) and consists of a vector of length n with elements that are positive for each of the salient ranks and 0 otherwise. As an example, for a four-team tournament, if the organizer decided that only the correctly choosing the best team as the winner was important, then $m = \{1, 0, 0, 0\}$ (i.e. measuring effectivity). In an event such as the Olympics, the salient weights could be 1's for the first three elements and 0's elsewhere such that $m = \{1, 1, 1, 0\}$. Additionally, one can also weight the salient ranks. For example, continuing the Olmypics example where getting the top three correct is the main objective, one could argue that it is more important to get the first place correct than the third place. This could lead to, for instance, a salient weight vector of $m = \{5, 3, 1, 0, \cdots, 0\}$ giving 5 times the weight to getting the first place team correct as getting the third team correct.

With the addition of the salient weighting vector, m, the formula for weighted mutual information becomes:

$$I_w(r, \hat{r}, m) = \sum_r \sum_{\hat{r}} w(r, \hat{r}, m) p(r, \hat{r}) \log \frac{p(r, \hat{r})}{p(r)p(\hat{r})}$$

.

Incorporating m into this function is straightforward for many weighting functions. We use the weighted Kendall's τ presented in Shieh (1998) with a slight modification to the sign function. Specifically, in the special case that $sign(0) = 1$. Then $w(r, \hat{r}, m)$ becomes:

$$w(r, \hat{r}, m) = \frac{\sum_{i \neq j} m^{\star}_{ij} \text{sign}_{(0+)}(r_i - r_j) sign_{(0+)}(\hat{r}_i - \hat{r}_j)}{\sum_{i,j} m^{\star}_{i,j} - \sum_i m^{\star}_{i,i}}$$

where $m^{\star}_{i,j} = m_{min(i,j)}$ and $sign_{(0+)}(.)$ is the sign of the input with the special case that a 0 maps to 1.

Finally, because this mutual information is unitless, the formula can be standardized to be between 0 and 1 as follows:

$$I^{\star}_w(r, \hat{r}, m) = \frac{I_w(r, \hat{r}, m)}{H(r, \hat{r})}$$

where $H(.)$ is the usual Shannon entropy. Using this framework, values of $I^{\star}_w(r, \hat{r}, m)$ near 1 indicated competition structures with high efficacy. Values near 0.25 indicate tournament structures where all possible rankings are equally likely and values near 0 indicate rankings that are inefficacious (e.g. complete rank reversal).

Under the assumption of equal probability for all possible permutations of $\hat{r}$, we can simplify $p(r, \hat{r}) = \frac{1}{n!}$ and $\sum_r \sum_{\hat{r}} w(r, \hat{r}, m) = 0.5 \times n!$. Assuming that $p(r) = p(\hat{r}) = \frac{1}{n!}$, this results in:

$$I_w(r, \hat{r}, m) = \sum_r \sum_{\hat{r}} w(r, \hat{r}, m) p(r, \hat{r}) \log \frac{p(r, \hat{r})}{p(r)p(\hat{r})} = 0.5 \times n! \times \frac{1}{n!} \times \log(\frac{1/n!}{(1/n!)^2}) = 0.5 \times \log(n!)$$

Additionally, we can assume that r and $\hat{r}$ are independent and have equivalent probabilities of $\frac{1}{n!}$. Through the chain rule detailed in Cover and Thomas (2006),

$$H(r, \hat{r}) = H(r) + H(\hat{r}|r)$$

. Assuming independence and equality,

$$H(\hat{r}|r) = H(\hat{r}) = H(r)$$

. Thus,

$$H(r, \hat{r}) = 2 \times H(r) = -2 \sum_r p(r) \log(p(r)) = -2(n!) \times (\frac{1}{n!}) \times \log(\frac{1}{n!}) = \log((n!)^2)$$

and the final simplification:

$$I_w^\star(r, \hat{r}, m) = \frac{I_w(r, \hat{r}, m)}{H(r, \hat{r})} = \frac{0.5 \times \log(n!)}{2 \times \log(n!)} = \frac{0.5}{2} = 0.25$$

5 Results

While we have explored many more competition scenarios, for the sake of space we present only results of tournaments with $n = 16$ with the distribution of true strengths distributed as equally spaced quantiles of a standard normal distribution unless otherwise noted. In order to simulate each game we compute the probability that team j will beat team k by taking the inverse logit transformation of the differences in their strength parameters such that

$$p_{jk} = \frac{e^{\theta_j - \theta_k}}{1 + e^{\theta_j - \theta_k}}$$

where p_{jk} is the probability that competitor j beats competitor k and θ_j and θ_k are the strength parameters of teams j and k, respectively.

We focus on several simple competition structures focused mainly on variations of bracket tournaments to demonstrate the proof of concept of our proposed metric. In all cases, where

seeding is needed (i.e. all the bracket structures), the seeds are considered to be known and correct reflecting each team's true rank. Note that in bracket structures, the final rankings of teams not reaching the finals was randomly assigned among possible ranks. For example, if a team in a single elimination bracket lost in the semi-finals, their possible ranks are either 3 or 4 and are equally likely; if a team loses in the quarterfinals in that same tournament, their final rank is randomly assigned from the set of ranks 5 through 8. Each competition structure was simulated 10,000 times and those results were used to estimate $wI_m(r, \hat{r})$.

We examined the following structures with the assumption of the ideal seeding for each competition:

- Single Elimination, seeding (1 vs 16, 8 vs 9, 4 vs 13, 5 vs 12, 3 vs 14, 6 vs 11, 7 vs 10, 2 vs 15)
- Double Elimination, seeding (1 vs 16, 8 vs 9, 4 vs 13, 5 vs 12, 3 vs 14, 6 vs 11, 7 vs 10, 2 vs 15)
- Round Robin
- Group Stage, 4 groups of 4, seeding (pot 1: 1-4, pot 2: 5-8, pot 3: 9-12, and pot 4: 13-16)
- Staged Round Robin, 4 groups of 4, seeding (pot 1: 1-4, pot 2: 5-8, pot 3: 9-12, and pot 4: 13-16)
- Tennis Ladder, seeding (16 vs 15, 16 vs 15 winner vs 14, etc.)

Figure 1 shows the results of the estimated values of $wI_m(r, \hat{r})$ on the y-axis while the x-axis represents the number of salient weights that were considered. Intuitively, of the competition structures considered here, the tennis ladder (shown in brown) should be the best structures because its built in structure limits the amount the top teams can fall in the final $\hat{r}$ rankings. Note that it tops almost all of the bracket structures regardless of the choice of the number of salient ranks and distribution of θ The exception comes in that round robin has the largest number of matchups and, with the exponential distribution, the probability of the top seed winning each game is much larger than the other distributions.

The distribution of θ can highly influence the resulting weighted mutual information that results. If the distribution is "Coin Flip", as in every team has a 50% chance to win each game, then the overall information that can be gathered from the competition structure itself is significantly lower than other distributions. Additionally, the exponential distribution shows a larger gap between single elimination (shown in red) and double elimination (shown in blue) than the normal distribution, when k is small. This is because the probability of the better team winning is increased, especially for the top few teams.

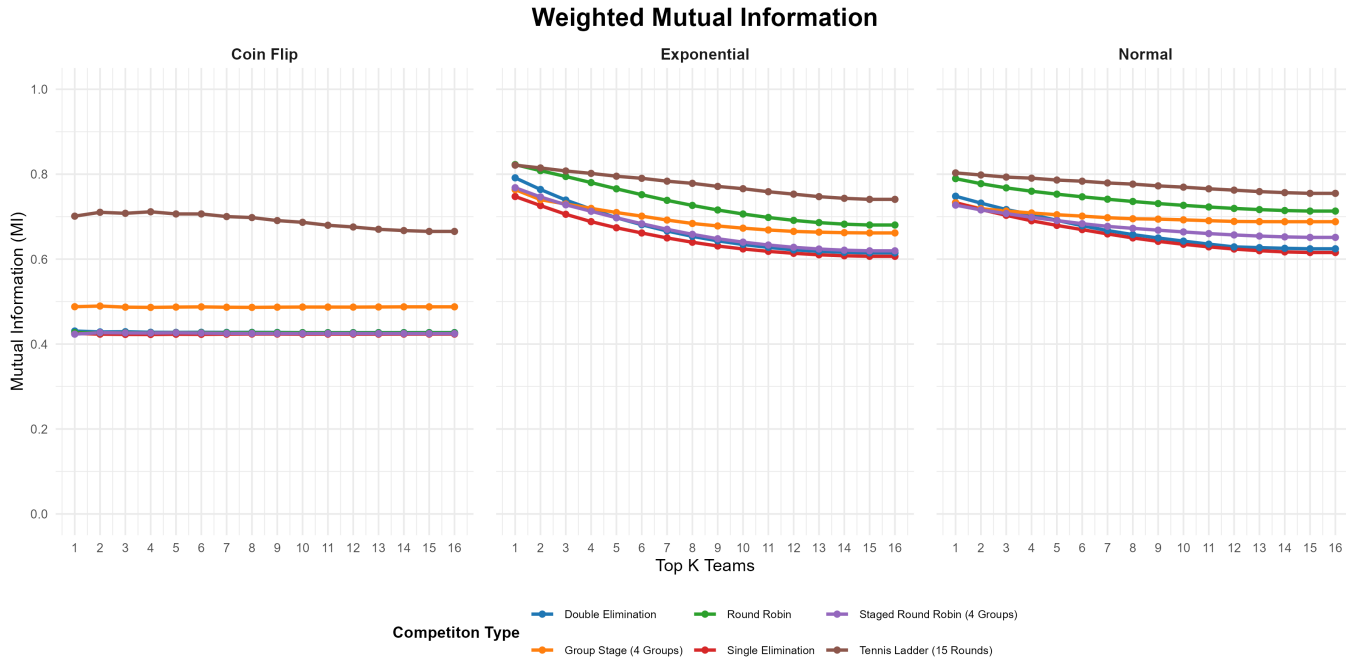

Figure 1: This figure compares the normalized weight mutual information of various competition structures against the number of salient ranks for various assumed distributions of true strengths, θ.

It is important to note that while the theoretical value for round robin for the distribution "Coin Flip" is 0.25, it is not seen in the results due to a limited number of simulations and noise within simulations.

In addition to the estimated values of $wI_m(r, \hat{r})$, we can also easily estimate the tiered rankings.

Figure 2 demonstrates tiered ranking for the top K teams. As the top K teams approach the total number of teams, the mutual information approaches 1, or complete information. When K is 16, every team is within the top 16, so it is correctly ordered.

While the comparison between most of the structures show similar results to those in Figure 1, the plot of Group Stage shows a noticeable different with a its unique pattern. This is due to the ranking within the structure, as within each group, the pots that determine the groups create groups with one team from the top 4 teams, then one team from the next 4, etc. This creates the sharp inclines seen from $K = 11$ to $K = 12$ and $K = 15$ to $K = 16$. If one of the bottom 4 teams ends up in third place or higher in the group phase, then the mutual information lowers. The random assignment of ranks within the teams that finish in last create the plateaus because, given the lowest 4 teams all end up last in their respective groups, those teams should have an equal probability of ending up with rank 13 as rank 16.

In order to avoid these leaps, a potential solution would be to create multiple tiers: the top 8

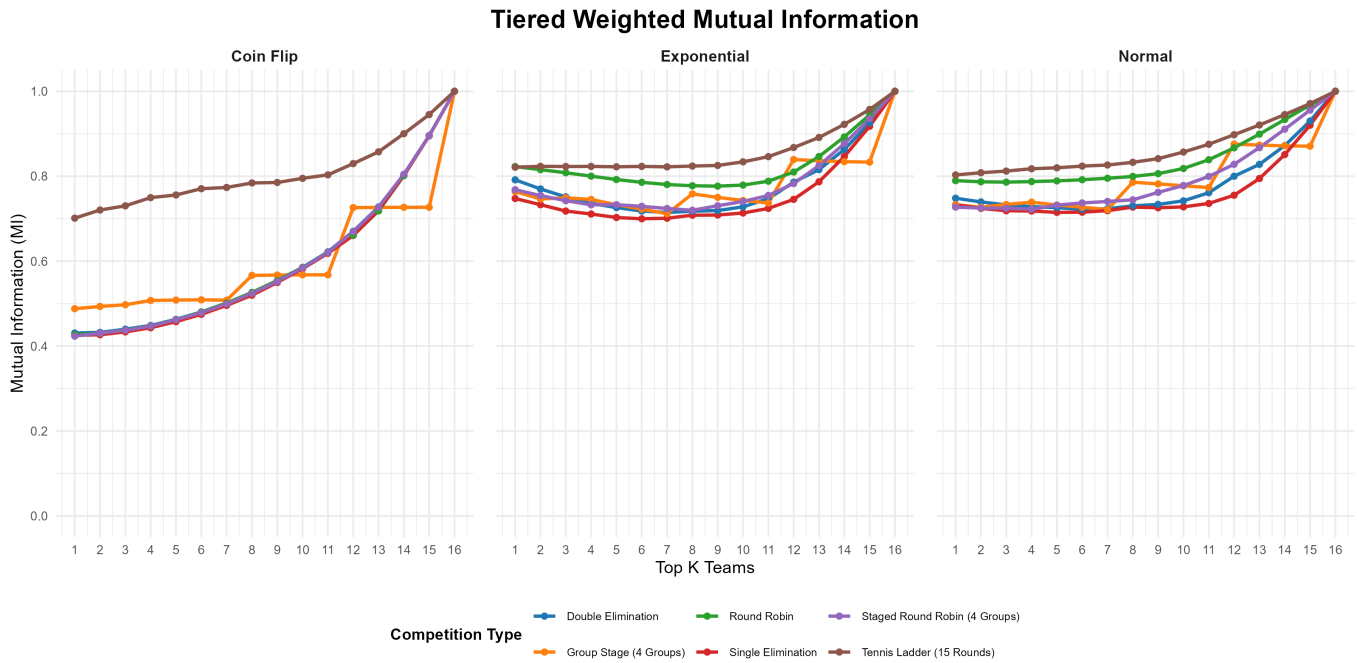

Figure 2: This figure compares the normalized weight mutual information of various competition structures against the number of *tiered ranks* for various assumed distributions of true strengths, θ.

teams (or teams that advance to the knockout round), the teams that finished third in their respective groups, and the teams that finished last in their respective groups. However, for the sake of comparing against other structures, we only looked at the top K teams.

In addition to the unique competition structures, most structures have different variations of the overall structure. For example, single elimination tournaments could have a reseeding structure, like the NFL playoffs. In this case, instead of set matchups, the matchups are determined as the highest seed remaining plays the lowest seed remaining, then the next highest plays the next lowest until all teams have a game.

Another structure is the step Ladder structure in which the lowest seed plays the next lowest seed (16 vs 15) and the winner of that plays the next lowest (14). This continues until the top team plays a game. If this is correctly seeded, it seems like the ideal structure with very minimal variation. However, if seeded poorly, the structure could be viewed as one of the worst. Therefore, Figure 3 below looks at different structures, seeding, and number of games within a series.

Note, the labelled "Bad Seed" is equivalent to the worst possible seeding structure. For step ladder, it means the top team plays the second best team, the winner plays the third best team etc. For single elimination, the seeding structure is (1 vs 2, 3 vs 4, 5 vs 6, 7 vs 8, 9 vs 10, 11 vs 12, 13 vs 14, 15 vs 16).

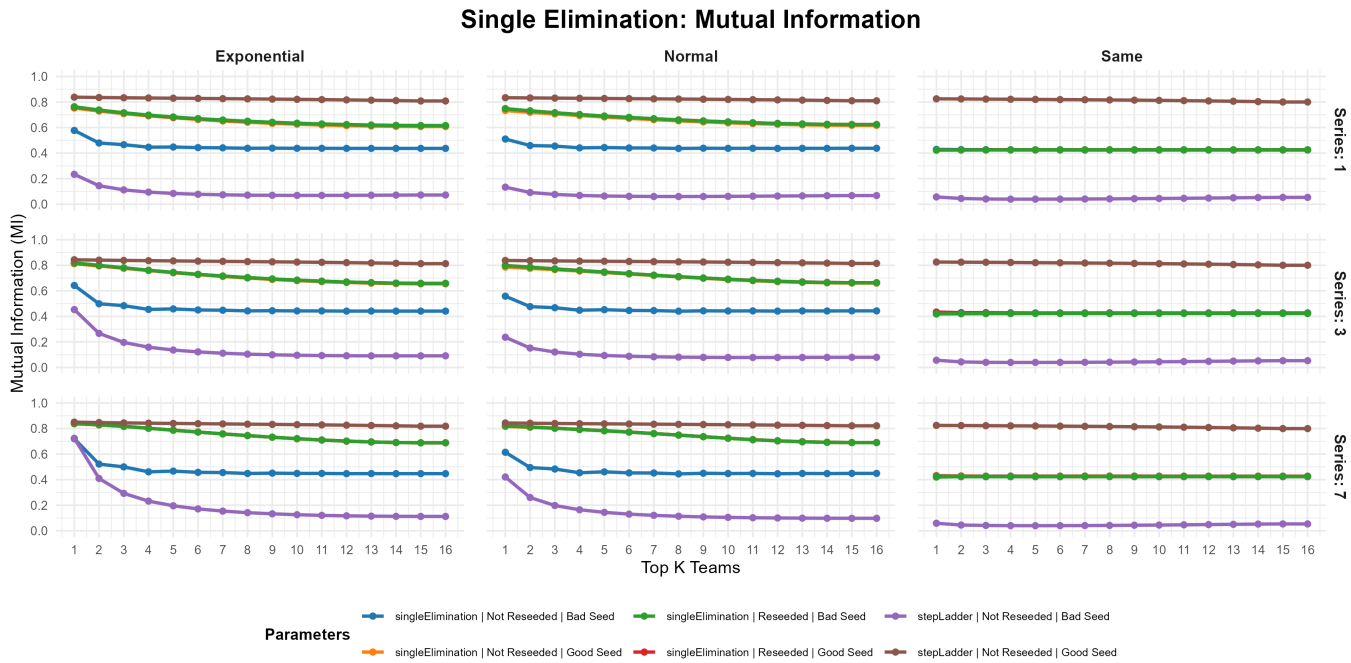

Figure 3: This figure compares the normalized weight mutual information of various single elimination competition structures against the number of *tiered ranks* for various assumed distributions of true strengths, θ.

Once again, the best and worst structure are the expected of step ladder, good and bad seeding structures. The step ladder with the bad seeding structure actually is below 0.25, or completely random assignment of $\hat{r}$, for larger values of K. The distribution, however, heavily influences the resulting mutual information in the smaller values of K for the step ladder with bad seeding, but not as much with good seeding. This can be attributed to the fact, for good seeding structure, the top team has to win only 1 match, with the next be playing 2 matches, etc., while the top teams are still favored. However, for the bad seeding structure, the top team has to play 15 matches to end up in the correct position, and for that to occur, the second best team ends up in last place. Therefore, given the distribution, the probability that the best team wins 14 games can change the metric drastically. The same logic applies for the number of games in the series, as more games lead to higher mutual information. While still having an effect, the distribution plays less of factor in the typical single elimination structure.

Where the seeding structure heavily impacted the step ladder structure, the single elimination structure is seen to be less influenced by seeding, given that there is reseeding. As seen in Figure 3, the reseeded single elimination structure has very little difference between the good seeding and bad seeding. Additionally, both reseeded actually appear slightly better than the good seeding structure for the usual structure with a set bracket. This would lead us to believe that, when determining a structure to use, the reseeded option is a more

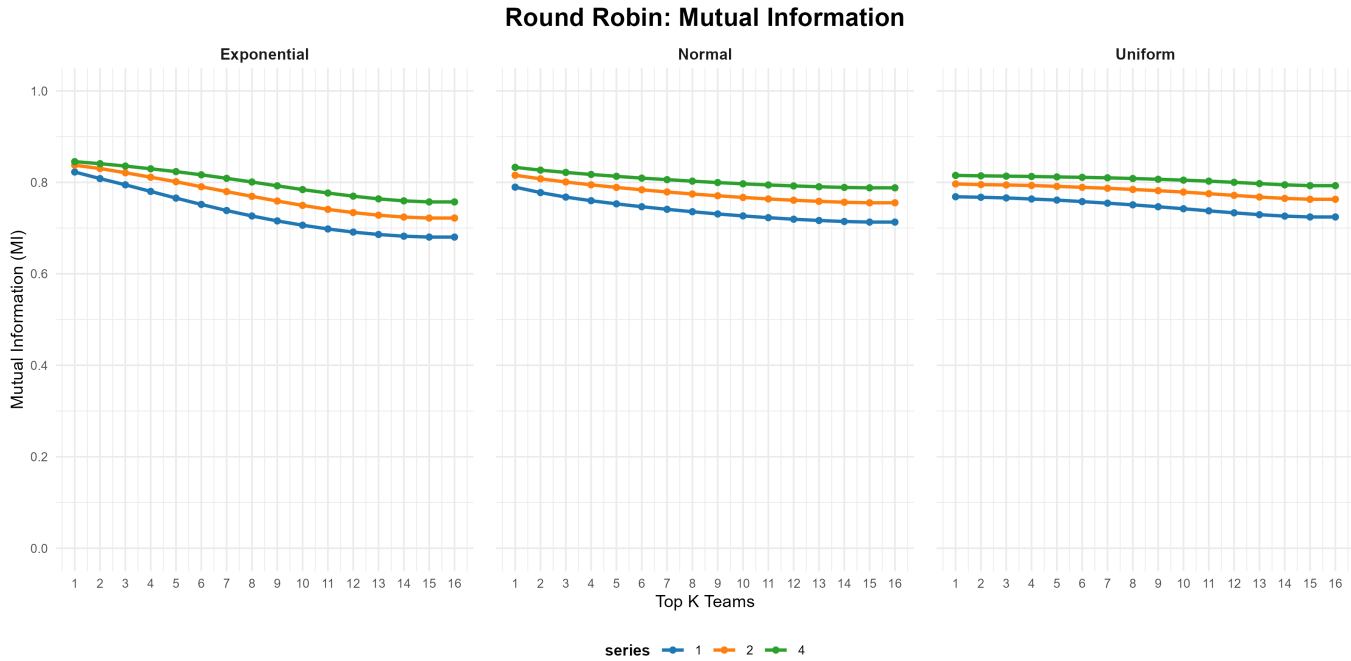

Figure 4: This figure compares the normalized weight mutual information of various round robin competition structures against the number of salient ranks for various assumed distributions of true strengths, θ.

efficacious approach than the common single elimination structure.

Another competition structure that can be used as a simple example is round robin. When time and number of games is not a constraint, this is typically the most common approach to a structure. Examples of this are found most often in European soccer leagues, like the Premier League.

Figure 4 depicts three different distributions and the mutual information given from a round robin structure for a number of series. As the number of rounds increase, so does the mutual information given that there is a difference in the strength parameters. As the number of rounds increase, the probability of the better team being ranked higher than a lower team increases. Therefore, given an infinite number of rounds, the metric should approach a value of 1.

It is also interesting to note that the exponential distribution has a wider gap between the curves as the value of K increases. This is likely because the difference between the strength parameters for the top 2 teams is larger than the difference between the strength parameters for the bottom 2. Therefore, the larger difference leads to less variation in games for the top teams and more variation for lower teams.

On the other hand, the uniform distribution only has a slight variation in the around the middle ($K = 8$) that appears to become more flat as the number of rounds increase to 4.

This is because the strength parameters are equally spaced, and thus have roughly the same variation from simulation to simulation. Over time, as the number of rounds increase, all of the curves will begin to flatten.

In a perfect world, a large number of rounds for a round robin tournament would always be used and all of the other structures would, thus, become useless. However, we do not live in a perfect world and athletes cannot sustain a season that long. So, how can this metric be able to find the best structure given the real-life constraints? One might suggest estimating the strengths of each competitor in a given season and using the structures that would be reasonable to use given the time constraints. This, however, would give the best competition structure for that specific season.

To estimate our metric for a specific real-world competition, some potential pathways include:

- estimating the strength parameters for many seasons, running simulations for each of those seasons, combining the results and using the formula to get an overall mutual information score
- estimating the strength parameters for many seasons, running simulations for each of those seasons and calculating the mutual information for each season, average the resulting metrics
- estimating the strength parameters for many seasons and averaging the strength parameter associate by the order of competitors (e.g. average the best team from each season), running simulations over the new strength distribution from the averages.

Overall, the metric succeeds in comparing competition structures effectiveness based on efficacy and effectivity. There is no one perfect structure that can be used given real-world constraints, but this metric can offer potential solutions to structures that struggle to provide the results that are expected and desired.

6 Conclusion

This study introduced a weighted mutual information framework to evaluate the effectiveness of different competition formats in preserving the true underlying rankings of teams. We also introduce the concept of salient ranks, thereby extending and generalizing the concepts of efficacy and effectivity. By simulating outcomes of competition structures with varying seeding methods and applying an information-theoretic approach, we quantified how accurately each format conveys ranking information. Our findings confirm the intuitive advantage of round robin tournaments, which consistently outperformed single and double elimination formats

in ranking accuracy, but also quantify the difference between these structures. Additionally, we demonstrated the degree to which poor seeding structures can significantly degrade performance of a competition structure, in some cases making competition outcomes even worse than simply randomly generating a set of ranks.

These results underscore the trade-off between tournament accuracy and practical constraints such as time, cost, and entertainment value. While round robin offers the highest fidelity to true rankings, it is often impractical for large tournaments, highlighting the need for hybrid or adaptive structures that balance accuracy with logistical feasibility.

Future work should extend this analysis to scenarios with more tournament structures, varying numbers of competitors, incorporate probabilistic strength models reflecting real-world uncertainty, and explore alternative weighting schemes for different competitive priorities (e.g., 4 salient ranks versus 8 salient ranking). Applying these methods to actual tournament data could further validate their usefulness for organizers aiming to design fair and informative competitions.

Acknowledgements

We thank the Department of Mathematics and Statistics at Loyola University Chicago for their support and resources in conducting this study. We thank Kailey Marie Lum for the suggestion of the term "salient ranks". No external funding was received for this research.

Supplementary Material

All supplementary material available at https://github.com/gjm112/tournaments.

7 References

Appleton, David R. 1995. "May the Best Man Win?" *Statistician* 44 (4): 529.

Cover, Thomas M., and Joy A. Thomas. 2006. *Elements of Information Theory.* 2nd ed. Wiley-Interscience.

Csato, L. 2021. *Tournament Design: How Operations Research Can Improve Sports Rules. 1st Ed.* Cham, Switzerland: Palgrave Macmillan.

Daniels, H. E. 1969. "Round-Robin Tournament Scores." *Biometrika* 56 (2): 295–99. www.jstor.org.

Devriesere, Csató, K., and D. Goossens. 2025. "Tournament Design: A Review from an Operational Research Perspective." *European Journal of Operational Research* 324 (1): 1–21.

Glenn, W A. 1960. "A Comparison of the Effectiveness of Tournaments." *Biometrika* 47 (3-4): 253–62.

Guiasu, Silviu. 1977. *Information Theory with Applications*. New York, NY: McGraw-Hill.

Johnson, Sidney, and Rodney Fort. 2022. "Match Outcome Uncertainty and Sports Fan Demand: An Agnostic Review and the Standard Economic Theory of Ports Leagues." *Int. J. Emp. Econ.* 01 (02).

Lasek, Jan, and Marek Gagolewski. 2018. "The Efficacy of League Formats in Ranking Teams." *Stat. Modelling* 18 (5-6): 411–35.

Shieh, Grace S. 1998. "A Weighted Kendall's Tau Statistic." *Stat. Probab. Lett.* 39 (1): 17–24.

Sziklai, Balázs R, Péter Biró, and László Csató. 2022. "The Efficacy of Tournament Designs." *Comput. Oper. Res.* 144 (105821): 105821.